Daniel R. Okafor

Electrical & Electronic Engineering Graduate | d.okafor@aldermere.edu | github.com/drokafor

EDUCATION

M.Eng., Electrical Engineering — Aldermere Institute of Technology, 2021–2024
Thesis: Adaptive MPPT control for heterogeneous solar arrays. GPA 3.9/4.0.

B.Eng., Electrical & Electronic Engineering — Northgate University, 2017–2021
First-class honours, Power Electronics specialisation.

RESEARCH EXPERIENCE

Graduate Researcher, Power Systems Lab, Aldermere — 2022–2024
Grid-scale battery SOC estimation with adaptive EKF; 1.8% RMS error over 412 cycles.

Research Assistant, Embedded Sensing Group — 2023
Designed a sub-1 GHz sensor node with 14-month battery life; 12-node campus deployment.

Undergraduate Researcher, Northgate University — 2021
Wavelet-based MV fault classifier; 96% recall on 20k simulated events.

PUBLICATIONS

[1] Okafor, D. R. & Lindqvist, M. (2024). Adaptive MPPT for Heterogeneous Solar Arrays. *J. of Power Systems*, 41(3).

[2] Okafor, D. R. (2023). Wavelet Transient Classification for MV Faults. *IEEE Regional Conf. Proceedings*.

TECHNICAL SKILLS

Hardware: PCB design (KiCad), FPGA (VHDL), power electronics, oscilloscope & VNA measurement

Software: MATLAB/Simulink, Python (NumPy, SciPy), C/C++, PyTorch, Docker, Git

Methods: Control theory, signal processing, statistical modelling, finite-element simulation